

Makendy Midouin

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EDUCATION

New York University, Master of Science in Computer Science | GPA: 3.45 May 2025

- **NYU Entrepreneurial Institute:** Founded COLEAGUE; conducted 30+ user interviews to pivot technical architecture toward identified market needs
- **Relevant Coursework:** Algorithms, Database Systems, Big Data (Spark/Hadoop), AI, Software Engineering

St. Thomas Aquinas College, Bachelor of Science in Computer Science | GPA: 3.68 May 2022

SKILLS

Languages: Python, JavaScript/TypeScript, Node.js, Java, SQL, C#

Cloud & DevOps: AWS (Lambda, API Gateway, S3, DynamoDB, CloudWatch), Docker, GitOps (GitHub Actions, Travis CI), Infrastructure as Code (IaC), Multi-Environment Deployment

Monitoring & Data: Sentry, CloudWatch Logs, PostgreSQL, MongoDB, MySQL

Frameworks: React, Next.js, Express, Django, Flask

PROFESSIONAL EXPERIENCE

Adjunct Professor | *St. Thomas Aquinas College* Aug 2024 – Dec 2025

- **Mentoring Production-Ready Development:** Led 40+ students through the full software development lifecycle, emphasizing Agile methodologies, automated testing, and peer code reviews
- **Establishing Engineering Standards:** Designed curriculum for Cloud Computing (AWS) and CI/CD, focusing on the development of reliable, testable, and maintainable software systems.

Software Engineer | *Roadcall.co* June 2023 – Dec 2023

- **Improving Platform Reliability:** Engineered an admin communication feature using React Hooks and Redux, streamlining workflows for thousands of users while maintaining high architectural standards.
- **Full-Cycle Software Development:** Participated in all aspects of development including design, coding, testing, and bug fixing to maintain a high-availability portal handling 5,000+ daily submissions
- **Managing Multi-Environment Deployments:** Orchestrated deployment cycles across development, testing, and production environments, implementing CI/CD pipelines that eliminated 10+ legacy defects and ensured data consistency

Software Engineer | *Treehouse Strategy* May 2021 – Sept 2021

- **Optimizing System Performance:** Architected backend API and caching refactors, achieving a 40% reduction in load times for creative agency portals
- **Building Scalable UI Components:** Developed responsive data tables and reusable React components, significantly increasing client engagement through improved data insights and application scalability

XLAB Researcher | *St. Thomas Aquinas College* Sept 2019 – May 2022

- **Complex Systems Development:** Developed a VR application (Unity/C#) with proximity-based dialogue systems and motion capture; delivered a technical keynote that secured \$125,000 in STEM scholarships
- **Technical Leadership:** Led a team of 5 researchers using Agile/Scrum to coordinate development sprints and meet high-stakes national deadlines for complex, data-driven simulations

PROJECTS

SmartFRQ – AI Learning Platform | Full Stack Developer [Link](#)

- **ML Service Deployment:** Deployed AI services integrating OpenAI GPT-4o via serverless AWS Lambda with CloudWatch monitoring, API Gateway routing, and Docker containerization, ensuring reliable ML execution for 100+ practice sessions serving 10+ users
- **Utilized:** Python, Next.js, TypeScript, Supabase, OpenAI GPT-4o, AWS Lambda, API Gateway, CloudWatch, Docker

Stack Overflow Big Data Dashboard | Full Stack Developer [GitHub](#)

- **Cloud Infrastructure:** Transformed 100GB of StackOverflow data into interactive dashboard using MongoDB Atlas and AWS DynamoDB caching, delivering sub-second insights through strategic indexing and performance optimization
- **Utilized:** Python, MongoDB Atlas, scikit-learn, NLTK, Pandas, AWS DynamoDB, Next.js, Apache ECharts

Java Lambda Runner | Full Stack Developer [GitHub](#)

- **Serverless Infrastructure:** Deployed production-ready AWS Lambda microservice that automated isolated execution environments, eliminating manual setup for engineers and supporting high-volume secure code execution
- **Utilized:** Java, Node.js, Docker, AWS Lambda, API Gateway

LEADERSHIP EXPERIENCE

Brains & Motion Education – Site Director & Lead Coding Instructor Summer 2024 & 2025

- **Directing STEM Operations:** Managed product strategy and facility operations for 160+ students, ensuring smooth technical implementation and stakeholder relations